

1. Relational Model Concepts

The **Relational Model** is a database model proposed by **E. F. Codd**. It organizes data into tables (relations).

Main Concepts:

- **Relation:** A table (e.g., Student table)
- **Tuple:** A row in the table (record)
- **Attribute:** A column (field)
- **Domain:** Set of valid values for an attribute
- **Schema:** Structure of a relation

Example:

Student(sid, sname, city)
→ sid = attribute, row = tuple

2. Relational Model Notations and Characteristics

Notations:

Relation: $R(A_1, A_2, A_3 \dots)$

Example: Student(sid, name, city)

Tuple: $t \in R$

Cardinality: Number of rows

Degree: Number of attributes

Characteristics:

No duplicate tuples

Order of rows and columns does not matter

Each attribute has atomic values

Each relation has a unique name

Data is stored in tabular form

3. Keys in DBMS

Keys are used to uniquely identify records.

Types of Keys:

Super Key:

Any set of attributes that uniquely identifies a tuple

Example: (sid), (sid + name)

Candidate Key:

Minimal super key (no extra attributes)

Example: sid

Primary Key:

Selected candidate key used for identification

Example: sid

Alternate Key:

Candidate keys not chosen as primary key

Foreign Key:

Attribute that refers to primary key of another table

Example: deptid in Student table

4. Update, Delete and Insert Operations

These are basic operations to modify data.

Insert:

Adds new tuple

Example: Add new student

- Must follow domain and key constraints

Update:

Modifies existing data

Example: Change city of a student

- Should not violate constraints

Delete:

Removes tuple

Example: Delete a student record

5. Relational Algebra & Its Properties

Relational Algebra:

It is a procedural query language used to retrieve data from relations.

Basic Operations:

Selection (σ)

Projection (π)

Union ($\cup$)

Intersection ($\cap$)

Difference ($-$)

Join ($\bowtie$)

Cartesian Product ($\times$)

Properties:

Closed system (output is always a relation)

Based on set theory

No duplicate tuples

Order of tuples does not matter

Operations can be combined

-

Q6 Entity Integrity, Referential Integrity and Foreign Keys

Answer:

1. Entity Integrity:

Entity integrity ensures that every table has a **primary key**, and this key must not contain NULL values. It uniquely identifies each record.

Example:

Student(ID, Name) $\rightarrow$ ID cannot be NULL

2. Referential Integrity:

Referential integrity ensures consistency between related tables. It ensures that a foreign key value must exist in the referenced table.

Example:

Student(DeptID) must match Department(DeptID)

3. Foreign Key:

A **foreign key** is an attribute in one table that refers to the primary key of another table.

Example:

Student(DeptID) $\rightarrow$ references Department(DeptID)

Key Points:

- Prevents invalid data
- Maintains relationships between tables
- Avoids orphan records

Q. 7SELECTION and PROJECTION Unary Relational Operations

Answer:

In relational algebra, unary operations operate on a single relation. Two important unary operations are Selection and Projection.

1. Selection (σ):

Selection retrieves rows that satisfy a specific condition.

Syntax:

$\Sigma(\text{condition})(\text{Relation})$

Example:

Student(Age > 18) $\rightarrow$ returns students older than 18

2. Projection (π):

Projection retrieves specific columns from a table and removes duplicates.

Syntax:

$\Pi(\text{attribute_list})(\text{Relation})$

Example:

$\Pi(\text{Name, Age})(\text{Student})$

Key Points:

Selection filters rows

Projection selects columns

Projection removes duplicate values

Both operations improve query efficiency

Q8. Queries on Given Schema

Given:

Student (sid, sname, gender, city)

Department (deptid, deptname)

Course (cid, ename)

Answer:

These queries are expressed using **Relational Algebra**:

1. **Students living in Ahmedabad:**
 $\sigma(\text{city} = \text{"Ahmedabad"}) (\text{Student})$
→ Retrieves all tuples where city is Ahmedabad
 $\pi(\text{sname})(\sigma(\text{city} = \text{"Ahmedabad"}) (\text{Student}))$
2. **Female students:**
 $\sigma(\text{gender} = \text{"Female"}) (\text{Student})$
3. **Names of all departments:**
 $\pi(\text{deptname})(\text{Department})$
4. **List all courses offered:**
 $\pi(\text{ename})(\text{Course})$
5. **Students not from Delhi:**
 $\sigma(\text{city} \neq \text{"Delhi"}) (\text{Student})$

6. **IDs and names of male students:**

$\pi(\text{sid}, \text{sname})(\sigma(\text{gender} = \text{"Male"})(\text{Student}))$

7. **Students from Trivandrum or Bangalore:**

$\sigma(\text{city} = \text{"Trivandrum"} \text{ OR } \text{city} = \text{"Bangalore"})(\text{Student})$

Explanation:

- **Selection (σ)** filters rows based on condition
- **Projection (π)** selects specific columns

Q9 Various Types of Binary Relational Operations

Answer:

Binary operations operate on two relations. Important types include:

1. Union ($\cup$):

Combines tuples of two relations (no duplicates)

Example: $R \cup S$

2. Intersection ($\cap$):

Returns common tuples

Example: $R \cap S$

3. Difference ($-$):

Tuples in one relation but not the other

Example: $R - S$

4. Cartesian Product ($\times$):

Combines all tuples of R with all tuples of S

Example: $R \times S$

5. Join ($\bowtie$):

Combines related tuples based on condition

Example: Student $\bowtie$ Department

6. Division ($\div$):

Finds tuples related to all tuples in another relation

Q10 Aggregate Functions and Grouping in Relational Algebra

Answer:

Aggregate functions perform calculations on groups of tuples and return a single value.

Common Aggregate Functions:

COUNT() $\rightarrow$ number of tuples

SUM() $\rightarrow$ total values

AVG() $\rightarrow$ average

MIN() $\rightarrow$ smallest value

MAX() $\rightarrow$ largest value

Grouping Concept:

Grouping divides tuples into sets based on attribute values.

Example:

Student(Dept, Marks)

Group by Dept:

Dept A $\rightarrow$ AVG(Marks)

Dept B $\rightarrow$ AVG(Marks)

Relational Algebra Representation:

Γ Dept, AVG(Marks)(Student)

Q.11 (b) Key Concepts of Relational Model

Answer:

The relational model organizes data into structured formats called relations (tables).

1. Relation:

A table consisting of rows and columns

Example: Student table

2. Tuple:

A row in a relation representing a record

Example: (1, Amit, Male, Ahmedabad)

3. Attribute:

A column representing a property

Example: sid, sname, city

4. Domain:

Set of valid values for an attribute

Example: Gender → Male/Female

5. Keys:

Primary Key → uniquely identifies tuple

Foreign Key → links relations

6. Schema:

Structure of relation (attribute names and types)